

1. Administrative & Study Identification

Posting ID: 120044305868735283 **General Study Title:** A Study of Pembrolizumab (MK-3475) With or Without Intismeran Autogene (V940) in Participants With Non-small Cell Lung Cancer (V940-009/INTERpath-009) **Official Full Study Title:** A Phase 3 Randomized Double-blind Study of Adjuvant Pembrolizumab With or Without V940 in Participants With Resectable Stage II to IIIB (N2) NSCLC Not Achieving pCR After Receiving Neoadjuvant Pembrolizumab With Platinum-based Doublet Chemotherapy (INTERpath-009) **Short Title/Acronym:** V940-009 / INTERpath-009 **Protocol Number:** V940-009 **NCT Number:** NCT06623422 **Sponsor & Company Name:** Merck (Merck Sharp & Dohme LLC in collaboration with Moderna, Inc.) **Investigational Product:** Pembrolizumab (MK-3475) and Intismeran autogene (V940 / mRNA-4157) **Phase of Development:** Phase 3 (Criteria Type: PHASE_REQUIREMENT - Trial must be in PHASE3) **Trial Status:** RECRUITING (Currently recruiting participants) **Medical Monitor:** Merck Sharp & Dohme LLC Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate if adjuvant individualized neoantigen therapy V940 (intismeran autogene) combined with pembrolizumab improves disease-free survival (DFS) compared to placebo plus pembrolizumab in participants with NSCLC. **Secondary Objectives:** To evaluate overall survival (OS), distant metastasis-free survival (DMFS), and disease-free survival after next-line therapy (DFS2).

2.2 Background & Rationale Patients with non-small cell lung cancer (NSCLC) who do not achieve a pathological complete response (pCR) after neoadjuvant chemoimmunotherapy and surgery have a worse prognosis than those who achieve pCR. Escalation of therapy for improved outcomes may be warranted in this population. This study investigates whether giving the individualized neoantigen therapy V940 alongside pembrolizumab as an adjuvant treatment can help prevent cancer from coming back and prolong survival.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized (1:1)

3.2 Planned Enrollment & Duration Targeted Enrollment: 680 participants **Number of Sites:** Global, Multicenter **Estimated Study Start:** 21-Oct-2024 **Estimated Primary Completion:** 16-May-2033

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Histologically/cytologically confirmed diagnosis of previously untreated and pathologically confirmed resectable clinical Stage II, IIIA, or IIIB (N2) non-small cell lung cancer (NSCLC) [AJCC 8th Edition]. 3. Did not achieve a pathological complete response (pCR) following completion of neoadjuvant chemotherapy and pembrolizumab followed by surgery. 4. Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1 within 7 days before the first dose. 5. Confirmation that epidermal growth factor receptor (EGFR)-directed therapy is not indicated as primary therapy (absence of tumor-activating EGFR mutations [e.g., DEL19 or L858R]). 6. HIV-infected participants must have well-controlled HIV on anti-retroviral therapy (ART). 7. Hepatitis B surface antigen (HBsAg) positive participants are eligible if receiving HBV antiviral therapy for at least 4 weeks with undetectable viral load. HCV participants are eligible if viral load is undetectable at screening.

4.2 Exclusion Criteria 1. Diagnosis of SCLC or, for mixed tumors, presence of small cell elements, or has a neuroendocrine tumor with large-cell components, a sarcomatoid carcinoma, or a pancoast tumor. 2. Documentation indicating presence of anaplastic lymphoma kinase (ALK) gene rearrangements. 3. Received prior therapy with an anti-PD-1, anti-PD-L1, or anti-PD-L2 agent (other than the protocol-specified neoadjuvant therapy), or with an agent directed to another stimulatory/co-inhibitory T-cell receptor (e.g., CTLA-4, OX-40, CD137). 4. Received prior neoadjuvant therapy for their current NSCLC diagnosis (other than what is specified). 5. Received prior systemic anticancer therapy including investigational agents other than specified in this protocol. 6. Received prior treatment with a cancer vaccine. 7. Received prior radiotherapy within 2 weeks of start of study intervention, or has radiation-related toxicities requiring corticosteroids. 8. Received a live or live-attenuated vaccine within 30 days before the first dose of study intervention.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Following surgical resection, participants will receive pembrolizumab 400 mg every 6 weeks for up to 7 cycles PLUS V940 (intismeran autogene) 1 mg every 3 weeks for up to 9 doses. **Route of Administration:** Pembrolizumab via Intravenous (IV) infusion; V940 via Intramuscular (IM) injection. **Prior Protocol-Specified Therapy:** Neoadjuvant treatment involved pembrolizumab with platinum-based

doublet chemotherapy (e.g., involving **cisplatin**; Drug Preferred Name: cisplatin; Trade Name: Platinol; ATC Code: L01XA01).

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for study-related adverse events. **Prohibited:** EGFR-directed therapy, investigational agents other than specified in this protocol, other systemic anticancer therapies, and live/attenuated vaccines.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Tumor Imaging, Viral/Genetic Labs
Baseline	Day 0	Randomization, First Adjuvant Dose
Treatment	Q3W/Q6W	Pembrolizumab (Q6W) & V940 (Q3W) Administration, Safety Labs
Follow-up	Week 12 to Year 3+	Tumor imaging every 12 weeks until week 48, every 24 weeks through year 3, and every 48 weeks thereafter

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Disease-Free Survival (DFS): Defined as the time from randomization to any recurrence (local, locoregional, regional or distant), occurrence of new primary NSCLC, as assessed by the investigator, or death due to any cause, whichever occurs first.

7.2 Secondary & Exploratory Endpoints 1. Overall Survival (OS) 2. Distant Metastasis-Free Survival (DMFS) 3. Disease-Free Survival 2 (DFS2)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of adverse events during the treatment phase and throughout the specified safety follow-up period. **Serious Adverse Events (SAEs):** Expedited reporting to the Sponsor and Institutional Review Boards within 24 hours of site awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (DMC) will evaluate the safety and efficacy data throughout the trial.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy endpoints. Safety Set for all participants who receive at least one dose of study medication. **Sample Size Justification:** Target $N = 680$, powered to detect a statistically significant

improvement in DFS for the combination arm versus the placebo arm. **Handling of Missing Data:** Standard oncology conventions including censoring patients at the last date they were known to be alive and disease-free for time-to-event analyses.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Full audit trails through Electronic Data Capture (EDC) systems and data clarification forms.

11. Study Identifiers & Governance

Primary ID: {{V940-009}} **Registry Number:** {{NCT06623422}}

Sponsor/Lead Organization: {{Merck}} **Study Title:**

{{INTERpath-009}} **Official Regulatory Title:** {{A Phase 3 Randomized Double-blind Study of Adjuvant Pembrolizumab With or Without V940 in Participants With Resectable Stage II to IIIB (N2) NSCLC Not Achieving pCR After Receiving Neoadjuvant Pembrolizumab With Platinum-based Doublet Chemotherapy}}

12. Clinical Framework (NSCLC & Carcinoma Ontology)

Disease Areas: Carcinoma, Non-Small-Cell Lung **Preferred UMLS**

Name: Non-small cell lung cancer **Diagnosis Threshold:**

{{Resectable Stage II to IIIB (N2) NSCLC not achieving pCR}}

Medical Methodology: {{Adjuvant Individualized Neoantigen Therapy (INT) plus anti-PD-1 therapy}} **Optimization Target:** {{Prevention of disease recurrence and prolonged survival}}

Ontology Mapping (Carcinoma Specifics): * **Name / Preferred Name:**

Carcinoma * **Semantic Type:** Neoplastic Process * **Definition:** A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas (NCI Thesaurus). * **MeSH Code:** D002277 * **HPO Code:** HP:0030731 * **SNOMED ID:** 1187425009 * **Source Level:** 5

13. Study Procedures & Methodology

Management Pathway: {{Adjuvant treatment following neoadjuvant chemoimmunotherapy and surgical resection}} **Education Protocol (HCP):** {{Protocol-specific training on V940/Pembrolizumab

administration and immune-related AE management}} **Education Protocol (Patient):** {{Informed consent process, treatment schedule education, and side-effect reporting}} **Quality Audit Mechanism:** {{Clinical site monitoring, tumor imaging independent review, and safety reporting tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to ~129 months (Estimated Primary Completion in 2033)}} **Onsite Visit Cadence:** {{Every 3 weeks during V940 dosing, every 6 weeks during pembrolizumab dosing}} **Remote Monitoring Frequency:** {{As needed for safety and survival follow-up}} **Checklist Review Interval:** {{Tumor imaging every 12 weeks until week 48, every 24 weeks through year 3, and every 48 weeks thereafter}}

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator		[Lead Investigator Name]		_____		[DD- MMM-YYYY]			Clinical Operations Lead		[Merck Operations Lead Name]		_____		[DD-MMM-YYYY]	
										Lead Statistician		[Merck Statistician Name]		_____		[DD-MMM-YYYY]	